

Module Description

The Payment Management module is responsible for managing and recording the payment details associated with ticket bookings. It ensures that every booking has a corresponding payment record and maintains information such as payment amount, payment method, payment status, and transaction date. Since this project is developed for academic purposes, the payment process is simulated without integrating any real online payment gateway. The module provides a secure and organized way of storing payment information while allowing users to view their payment history and administrators to monitor all payment transactions.

The module begins when a user confirms a ticket booking. The system automatically calculates the total payable amount based on the ticket price and quantity selected. The user then proceeds to the payment page, where a payment method can be selected and the payment is simulated. Once the payment is successfully completed, a payment record is generated and linked to the corresponding booking. The booking status is updated accordingly, and the user receives a payment confirmation.

The Payment Management module stores all payment-related information in the Payment collection of the MongoDB database. Each payment record includes the payment ID, booking reference, payment amount, payment method, payment status, and payment date. This information can later be retrieved for displaying payment history, generating reports, or verifying booking transactions.

The administrator can access all payment records through the admin dashboard. This enables monitoring of completed transactions, verification of booking payments, and generation of summary reports for administrative purposes. Appropriate validation checks are performed to ensure that payment information is complete and correctly linked to an existing booking.

Objectives

- Simulate the ticket payment process.
- Generate payment records for every successful booking.
- Store payment details securely in the database.
- Maintain payment history for users.
- Allow administrators to view and manage payment records.
- Ensure every payment is associated with a valid booking.

Inputs

- Booking ID
- Payment Amount
- Payment Method
- Payment Status

Processing

- Validate booking details.
- Calculate the total payable amount.
- Generate a payment record.
- Store payment information in MongoDB.
- Update the booking status after successful payment.
- Display payment confirmation to the user.

Outputs

- Payment confirmation.
- Updated booking status.
- Payment history for users.
- Payment records for administrators.

Database Collection Used

Payment

Validation Rules

- Booking ID must exist.
- Payment amount must be greater than zero.
- Payment method cannot be empty.
- Payment status must be valid.
- Payment record should be created only once per booking.

Module Flow

1. User confirms ticket booking.
2. System calculates the total amount.
3. User selects a payment method.
4. Payment is simulated successfully.
5. Payment details are stored in the database.
6. Booking status is updated.
7. Payment confirmation is displayed.
8. Administrator can view payment records through the dashboard.

Expected Outcome

The Payment Management module provides an efficient and organized mechanism for recording ticket payments. It ensures that every successful booking has an associated payment record, maintains transaction history for users, and allows administrators to monitor payment information through the administrative dashboard. Although the payment process is simulated, the module accurately represents the workflow of an actual event ticket booking system and maintains consistency with the overall system architecture.